AFC–7617

Sub. Code
1BCEA4

U.G. DEGREE EXAMINATION, NOVEMBER 2015

Computer Science — Allied

PROGRAMMING IN C++

(CBCS – 2011 onwards)

Time : 3 Hours Maximum : 60 Marks

Part A (10 × 1.5 = 15)

Answer **all** questions.

1. What are the basic principles of Object Oriented Programming?

2. Define Token.

3. What are the characteristics of static data members?

4. Define Constructor.

5. Define Inheritance.

6. What is a Pure Virtual function?

7. What is the difference between Sequential access and Random access File?

8. What are the advantages of Template?

9. What do you mean by Exception handling?

10. What are the two types of Exceptions?

Part B (5 × 3 = 15)

Answer **all** questions, choosing either (a) or (b).

11. (a) Explain the Arithmetic expression and Relational expressions with example.

Or

(b) Explain do-while statement with an example.

12. (a) Explain with an example how an object can be passed as argument.

Or

(b) Explain the characteristics of a Destructor.

13. (a) Explain nesting of classes with an example.

Or

(b) Explain the formatted console I/O operations.

14. (a) Explain the following functions with example. (i) tellg( ) (ii) tellp( ) (iii) seekg( ).

Or

(b) Explain function Template with an example.

15. (a) Explain the exceptions in Constructors with an example.

Or

(b) Explain the exception handling constructs with an example.

Part C (3 × 10 = 30)

Answer any **three** questions.

16. Write a C++ program to find roots of a quadratic equation for all cases using OOP concept.

17. Write a C++ program to find the biggest two numbers in two different objects of two different classes using friend function.

18. Write a C++ program to calculate the total mark obtained by the students using Multiple Inheritance.

19. Write a C++ program to search a employee record in a file using his id number.

20. Explain the exceptions in class Template with a program.

AFC–7466

Sub. Code
1BCE2C1

B.Sc. DEGREE EXAMINATION, NOVEMBER 2015

Second Semester

Computer Science

PROGRAMMING IN C++ AND ALGORITHMS

(CBCS – 2011 onwards)

Time : 3 Hours Maximum : 75 Marks

Section A (10 × 2 = 20)

Answer **all** questions.

1. Define Token.

2. Define class.

3. What is static data member?

4. Define pointers.

5. What is constructor?

6. Define Destructor.

7. What is operator overloading?

8. What is purpose for inheritance?

9. What is the principle used in Insertion sort?

10. Define minimum spanning tree.

Section B (5 × 5 = 25)

Answer **all** questions, choosing either (a) or (b).

11. (a) Explain structure of C++ program.

Or

(b) Compare structure and class.

12. (a) How is memory allocated for objects? What is static member function?

Or

(b) How do pass object as function argument? Give an example.

13. (a) Explain multiple constructors in a class with an example.

Or

(b) Describe dynamic constructor with an example.

14. (a) Explain multilevel and multiple inheritance in C++.

Or

(b) How do binary operator overload with an example?

15. (a) Explain binary search algorithm with an example.

Or

(b) Discuss Prim's algorithm with an example.

Section C (3 × 10 = 30)

Answer any **three** questions.

16. Explain principles of object oriented programming.

17. Discuss friendly function with suitable example.

18. Explain copy constructor with example.

19. How polymorphism is implemented in C++?

20. Explain Dijkstra's algorithm.

AFC–7467

Sub. Code
1BCE3C1

B.Sc. DEGREE EXAMINATION, NOVEMBER 2015

Third Semester

Computer Science

DATABASE MANAGEMENT SYSTEMS

(CBCS – 2011 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. What is DBMS?

2. Describe the different levels of data abstraction.

3. What are the uses of functional dependencies?

4. Define temporal data.

5. Mention any two advantages of distributed systems.

6. What do you mean by intraquery parallelism?

7. How to create the Indexes in SQL?

8. Specify the types of user's privileges.

9. Define PL/SQL Trigger.

10. How do you create Cursor in PL/SQL?

Part B $(5 \times 5 = 25)$

Answer **all** questions by choosing either (a) or (b).

11. (a) What is a data model? List the types of data model used.

Or

(b) What is an Entity Relationship model? Explain with an example.

12. (a) Describe the features of good Relational Database designs.

Or

(b) Illustrate the processes of decomposition using functional dependencies.

13. (a) Explicate the concepts of Client-Server architecture.

Or

(b) List and explain the advantages of Distributed Databases.

14. (a) Discuss in detail about Data Integrity.

Or

(b) Define Synonym. Explain how to create Synonym with an example.

15. (a) How are Triggers and assertions defined in PL/SQL? Explain.

Or

(b) Explicate the aggregate functions in SQL with examples.

Part C $(3 \times 10 = 30)$

Answer any **three** questions.

16. Describe the various applications of DBMS.

17. Discuss the fourth normal form with an example.

18. Write a brief note on Parallel databases.

19. How do you create and maintain a table? Discuss with an example.

20. How packages are useful in PL/SQL? Give an example.

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AFC–7468

Sub. Code
1BCE4C1

B.Sc. DEGREE EXAMINATION, NOVEMBER 2015

Fourth Semester

Computer Science

JAVA PROGRAMMING

(CBCS – 2011 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A $(10 \times 2 = 20)$

Answer **all** questions.

1. What is a Browser?

2. Define Constant and Variable.

3. What do you mean by operator precedence?

4. Find the value of X? if X= $(1011)_2$ and $(1010)_2$.

5. Define Class and Object.

6. What are the applications of wrapper classes?

7. Define Exception.

8. Define Package.

9. What is an Applet?

10. Write the syntax and an example of the method which is used to draw an arc.

Part B $(5 \times 5 = 25)$

Answer **all** questions choosing either (a) or (b).

11. (a) Explain the features of java.

Or

(b) Explain the Java Virtual Machine.

12. (a) Explain nested — if statement with an example.

Or

(b) Write a Java program to print the Prime numbers between 400 and 700 using while loop.

13. (a) Explain static members with a program.

Or

(b) Write a java program to define an interface Area. Implement the interface to calculate the area of a rectangle.

14. (a) Write a java program to create a package called employee. Import the employee package to display the details of the employee. [name, employee-id, height and weight.

Or

(b) Explain the Array IndexOutof Bound exception with an program.

15. (a) Explain with an example how will you add an Applet to HTML file.

Or

(b) Explain how will you use control loops in App lets.

Part C (3 × 10 = 30)

Answer any **three** questions.

16. Write a java program to sort the set of names using command line arguments.

17. Write a Java program to find the sum of the numbers that are divisible by 5 and not divisible by 10 between 200 and 350.

18. Write a Java program to find volume of a square, rectangle and circle by overloading the method volume.

19. Explain with a program, how will you create the Thread by implementing Runnable interface.

20. Write a Java program to draw 3 concentric circles with different colors.

AFC–7469

Sub. Code
1BCE5C1

B.Sc. DEGREE EXAMINATION, NOVEMBER 2015

Fifth Semester

Computer Science

OPERATING SYSTEMS

(CBCS – 2011 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. What is meant by Swp-in and Swap-out?

2. What do you mean by Fragmentation?

3. What is Deadlock?

4. Define Spooling.

5. What is Parallel Processing?

6. What is a Dedicated Device?

7. What are the functions of File manager?

8. What are the levels in File management system?

9. What is a Pipe?

10. What are the design Goals of Unix Operating System?

Part B (5 × 5 = 25)

Answer **all** questions by choosing either (a) or (b).

11. (a) Explain any types of Operating system.

Or

(b) Explain Demand paging technique with a diagram.

12. (a) Write short notes on Cache memory.

Or

(b) Explain the strategies for handling Deadlock.

13. (a) Explain Process Cooperation.

Or

(b) Explain the operation of Tightly coupled Multiprocessing system.

14. (a) Explain the NOS development.

Or

(b) Compare Network and Distributed Operating system.

15. (a) Explain the UNIX memory management system.

Or

(b) Explain the UNIX File management scheme.

Part C (3 × 10 = 30)

Answer any **three** questions.

16. Explain any Two Page replacement algorithms with example.

17. Explain Round Robin and FCFS scheduling algorithms with example.

18. Explain the components of the I/O subsystems.

19. Explain various File Access methods with diagram.

20. Explain the UNIX Device Management system.

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AFC–7470

Sub. Code
1BCE5C2

B.Sc. DEGREE EXAMINATION, NOVEMBER 2015

Fifth Semester

Computer Science

COMPUTER ORGANIZATION AND ARCHITECTURE

(CBCS – 2011 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. What are the uses of three state buffer?

2. Write the Micro operation for fetch cycle.

3. What is the function of STA Instruction?

4. What are the functions of CLE and CMA Instructions?

5. Define – Immediate Addressing mode.

6. What are the operations done with STACK?

7. Explain the need of Input-Output Interface.

8. Draw the flow chart for Multiplication Algorithm of Fixed Point numbers.

9. Write the differences between static RAM and Dynamic RAM.

10. What is meant by mapping in cache memory?

Part B $(5 \times 5 = 25)$

Answer **all** questions, choosing either (a) or (b).

11. (a) Explain the Register Reference Instructions.

Or

(b) Explain the Input-Output Instructions.

12. (a) Describe the Stages of Assembler.

Or

(b) Explain the types of Interrupts in detail.

13. (a) Explain the Data manipulation Instructions.

Or

(b) Explain the Single Accumulator Organization.

14. (a) Describe the Program Interrupt.

Or

(b) Explain the Booth's Multiplication Algorithm.

15. (a) Explain the Associative Memory logic.

Or

(b) Explain the Direct Mapping in Cache Memory.

Part C $(3 \times 10 = 30)$

Answer any **three** questions.

16. Describe the Instruction cycle in detail.

17. Explain about the machine language.

18. Explain the General Register Organization in detail.

19. Describe the Direct Memory Access data transfer.

20. Explain the Virtual Memory.

AFC–7471

Sub. Code
1BCEE1A

B.Sc. DEGREE EXAMINATION, NOVEMBER 2015

Fifth Semester

Computer Science

Elective — DATA MINING AND DATA WAREHOUSING

(CBCS – 2011 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. What is a data warehouse?

2. What is the need for backup?

3. What are the purpose of tools to manage a data warehouse?

4. What is process?

5. What is meant by KDD?

6. What are the metrics used in data mining?

7. Define Fuzzy set.

8. What is decision support system?

9. Define Itemset.

10. What is rule?

Part B $(5 \times 5 = 25)$

Answer **all** questions, choosing either (a) or (b).

11. (a) Explain the role of load manager in process architecture.

Or

(b) Explain the role of query manager in process architecture.

12. (a) Explain functions of warehouse manager process.

Or

(b) How do estimate the load in capacity planning?

13. (a) Discuss the social implications of data mining.

Or

(b) Explain data mining from a database perspective.

14. (a) Write short notes on web search engines.

Or

(b) Briefly discuss about OLAP.

15. (a) How do measure the quality of Rules? Explain.

Or

(b) Explain Basic Algorithm used in Association rules.

Part C (3 × 10 = 30)

Answer any **three** questions.

16. Discuss process flow within a data warehouse.

17. Explain about techniques for tuning queries.

18. Explain various issues in data mining.

19. Explain Genetic Algorithm as data mining technique.

20. Discuss advanced association rule techniques.

AFC–7472

Sub. Code
1BCEE1B

B.Sc. DEGREE EXAMINATION, NOVEMBER 2015

Fifth Semester

Computer Science – Elective

DIGITAL IMAGE PROCESSING

(CBCS – 2011 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. What is neighbours of a pixel?

2. What are the problems associated with visual perception?

3. Write continuous and discrete Fourier transform for one variable.

4. What is Fourier transform?

5. How do filter high frequencies?

6. What is image enhancement?

7. What is thresholding in image segmentation?

8. What is information gain?

9. What is close operation in morphology?

10. What is pattern classes?

Part B $(5 \times 5 = 25)$

Answer **all** questions, choosing either (a) or (b).

11. (a) Explain about sampling and quantization.

Or

(b) Briefly describe about imaging geometry.

12. (a) What is Fourier transform? Explain DFT for two variable.

Or

(b) Briefly describe about hotelling transform.

13. (a) What are the significance of colour image processing?

Or

(b) How enhancement is done in the frequency domain?

14. (a) Explain any one of the methods for boundary detection.

Or

(b) How to use motion in segmentation?

15. (a) Explain about elements of image analysis.

Or

(b) Discuss about structural methods.

Part C $(3 \times 10 = 30)$

Answer any **three** questions.

16. Discuss about fundamental steps in image processing.

17. Discuss role of FFT in image processing.

18. Explain least square restoration.

19. Discuss image compression models.

20. Explain about morphology operations.

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AFC–7473

Sub. Code
1BCEE2A

B.Sc. DEGREE EXAMINATION, NOVEMBER 2015

Fifth Semester

Computer Science – Elective

WEB DESIGN TECHNOLOGY

(CBCS – 2011 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. Name the types of markup languages.
2. Write the HTML code for create a drop down list.
3. Write the HTML code for insert a image in a web page.
4. What is meant by cascading style sheet?
5. Write the syntax for its structure in Java script.
6. Write the syntax for break statement in Java script.
7. Write the duration of identifiers in Java script.
8. What is meant by multiple subscripted array?
9. What is meant by event handlers?
10. What are the applications of XML?

Part B $(5 \times 5 = 25)$

Answer **all** questions, choosing either (a) or (b).

11. (a) How the size and style of a text is fixed in HTML?

Or

(b) Design a employment application form in HTML.

12. (a) Explain about the embedded style sheets.

Or

(b) Explain the positioning of elements in cascading style sheets.

13. (a) Explain the while structure with example in Java script.

Or

(b) What are the logical operators are used in Java script?

14. (a) Explain Date object in detail.

Or

(b) Explain about the math object in detail.

15. (a) Explain the XML namespaces with syntax.

Or

(b) Describe about XML vocabularies.

Part C (3 × 10 = 30)

Answer any **three** questions.

16. Explain the usage of <FRAMESET> tag with example.

17. Explain about the linking of external style sheets in detail.

18. Explain about the switch structure and do/while structure in Java script.

19. How will you declare and allocating arrays in Java Script?

20. Explain about the XML schema in detail.

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AFC–7474

Sub. Code
1BCEE2B

B.Sc. DEGREE EXAMINATION, NOVEMBER 2015

Fifth Semester

Computer Science

Elective — ADVANCED JAVA PROGRAMMING

(CBCS – 2011 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. Write any four character stream classes.

2. What is a stub?

3. What do you mean by Delegation Event model?

4. Write any two constructors of JButton.

5. What is FlowLayout?

6. How will you add an awt control to a window?

7. What is the purpose of InetAddress class?

8. What is URL?

9. What is a Java Bean?

10. What is a Cookie?

Part B $(5 \times 5 = 25)$

Answer **all** questions, choosing either (a) or (b).

11. (a) Explain with an example how will you write files.

Or

(b) Explain the methods of PrintReader class with example.

12. (a) Explain the constructors and methods of ItemEvent class with example.

Or

(b) Explain the constructors and methods of Scroll Panes with example.

13. (a) Explain the constructors and methods of CheckBox Group with example.

Or

(b) Write a java program to create a dialogBox in a Frame.

14. (a) Explain the functions of TCP/IP.

Or

(b) Explain Datagrams with a program.

15. (a) Explain Java Bean Bound and constrained properties.

Or

(b) Explain the life cycle of a servlet.

Part C (3 × 10 = 30)

Answer any **three** questions.

16. Explain the steps in RMI with an example.

17. Write a Java program to display your department particulars in JTable.

18. Write a java program to create two menu each menu contains three menuitems.

19. Explain the methods available in URL connection.

20. Explain handling HTTP POST requests with an example.

AFC–7475

Sub. Code
1BCE6C1

B.Sc. DEGREE EXAMINATION, NOVEMBER 2015

Sixth Semester

Computer Science

COMPUTER NETWORKS

(CBCS – 2011 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. Why do standardize network?

2. Define protocol.

3. What are the advantages of (any two) wireless transmission?

4. What is the main function of physical layer?

5. How do detect error using parity bit?

6. What is the disadvantage of stop and wait protocol?

7. What is meant by adaptive routing?

8. What is the advantage and disadvantage of connection less protocol?

9. What is cryptography?

10. What is WWW?

Part B $(5 \times 5 = 25)$

Answer **all** questions choosing either (a) or (b).

11. (a) Explain hardware and software required for network.

Or

(b) Explain uses of computer network.

12. (a) Write short notes on public switched telephone network.

Or

(b) Describe characteristics of any two guided transmission media.

13. (a) Explain CRC code with an example.

Or

(b) Describe sliding window protocol.

14. (a) What are the transport services offered by transport layer?

Or

(b) Explain about connection oriented transport protocol.

15. (a) Discuss the importance of DNS.

Or

(b) Explain working principle in E-mail.

Part C $(3 \times 10 = 30)$

Answer any **three** questions.

16. Explain ISO reference model with neat diagram.

17. Explain theoretical basis for data communication.

18. Explain multiple access protocol in Ethernet.

19. Discuss distance vector routing algorithm.

20. Explain public key algorithm.

AFC–7476

Sub. Code
1BCE6C2

B.Sc. DEGREE EXAMINATION, NOVEMBER 2015

Sixth Semester

Computer Science

COMPUTER GRAPHICS

(CBCS – 2011 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. Define Frame Buffer.

2. How will you generate Thick Line Segment?

3. What are the various Line styles?

4. What is a Concave Polygon?

5. What is Inverse Transformation?

6. How will you delete a segment?

7. Define Viewport.

8. What are conditions for a line clipping?

9. What are Sampled devices?

10. Define Event.

Part B $(5 \times 5 = 25)$

Answer **all** questions choosing either (a) or (b).

11. (a) Explain the operations on Computer graphics.

Or

(b) Write short notes on Antialiasing of lines.

12. (a) Explain the Display control with diagram.

Or

(b) Explain the Inside-Outside test in polygon with a diagram.

13. (a) Explain Segment Table with an example.

Or

(b) Write short notes on Raster Techniques.

14. (a) Explain the Viewing transformation.

Or

(b) Explain Sutherland Hodgman polygon clipping algorithm.

15. (a) Explain the steps in simulating a Locator.

Or

(b) Write short notes on Event Handling.

Part C $(3 \times 10 = 30)$

Answer any **three** questions.

16. Write and explain the Bresenham's line drawing algorithm.

17. Write and explain the Polygon interfacing algorithm

18. Describe the basic transformations with necessary diagram.

19. Write and explain line clipping with Cohen-Sutherland algorithm.

20. Explain the Interactive techniques.

AFC–7477

Sub. Code
1BCE6C3

B.Sc. DEGREE EXAMINATION, NOVEMBER 2015

Sixth Semester

Computer Science

SOFTWARE ENGINEERING

(CBCS – 2011 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. How will you define a "large project"?

2. How will you plan a "development process"?

3. Write the key principles of "Expert judgment" cost estimation.

4. What is meant by software requirements specification?

5. What is meant by abstraction?

6. What is meant by coupling and cohesion?

7. Define –acceptance testing.

8. Define unit testing.

9. What is meant by quality assurance?

10. Define –"Walk through".

Part B $(5 \times 5 = 25)$

Answer **all** questions, choosing either (a) or (b).

11. (a) What are the factors that influence quality and productivity? Explain.

Or

(b) What are the contents of an architectural design specification?

12. (a) Explain the staffing level estimation in detail.

Or

(b) Explain about the regular expressions.

13. (a) Explain about the HIPO diagrams.

Or

(b) Explain about the stepwise refinement.

14. (a) Describe about the configuration management in detail.

Or

(b) Describe about the integration testing in detail.

15. (a) How will you prepare a quality assurance plan?

Or

(b) Explain about the inspections.

Part C (3 × 10 = 30)

Answer any **three** questions.

16. Explain the managerial issues in detail.

17. Explain about the formal specification techniques.

18. Explain about the structural coding techniques.

19. Explain the managerial aspects of software engineering.

20. Describe about the software reviews in detail.

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AFC–7478

Sub. Code
1BCEE3A

B.Sc. DEGREE EXAMINATION, NOVEMBER 2015

Sixth Semester

Computer Science

Elective — MOBILE COMMUNICATION

(CBCS – 2011 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. What are the disturbances affect the wireless transmission?

2. Draw the frequency spectrum and mention the frequency Ranges.

3. Which principle is used in digital Audio Broad Casting?

4. What is meant by MEO 140?

5. What is meant by ADHOC networks?

6. What is meant by Quality of Service?

7. Define : Mobile TCP.

8. Define : Mobile IP.

9. Write the origin of HTML.

10. Write the HTML code for set the Background of a web page.

Part B $(5 \times 5 = 25)$

Answer **all** questions, choosing either (a) or (b).

11. (a) Describe about the cellular systems.

Or

(b) Describe about the signals used in Radio Transmission.

12. (a) Explain about TETRA.

Or

(b) Explain about DECT.

13. (a) Describe about the wireless LAN.

Or

(b) Describe about location management.

14. (a) Explain about snooping TCP.

Or

(b) Explain about the dynamic host configuration protocol.

15. (a) How special characters are used with the HTML?

Or

(b) Write the unique features of WAP.

Part C (3 × 10 = 30)

Answer any **three** questions.

16. Describe about the Antennas used in Wireless Transmission.

17. Explain about IMT – 2000 and LEO 139.

18. Describe about Access pointer control protocol.

19. Describe about transaction oriental TCP.

20. Describe about file systems consistency.

AFC–7479

Sub. Code
1BCEE3B

B.Sc. DEGREE EXAMINATION, NOVEMBER 2015

Sixth Semester

Computer Science

Elective – VISUAL BASIC

(CBCS – 2011 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. What is a constant?

2. Define function.

3. Write the form's components.

4. Write any four properties of combo box.

5. Write any two graphics control and its use.

6. What is current X and current Y properties?

7. What are the uses of Rich Text Box control?

8. Write the properties of List View control.

9. What is a data control?

10. Write any two SQL query.

Part B (5 × 5 = 25)

Answer **all** questions choosing either (a) or (b).

11. (a) Write the difference between arrays and collection.

Or

(b) Explain 'select case' statement with suitable example.

12. (a) How to built dynamic form at runtime give example.

Or

(b) Explain the basic properties and methods of list box control.

13. (a) Explain co-ordinate system with scale properties.

Or

(b) Explain optimization issues in graphics controls.

14. (a) How the tree structure work?

Or

(b) Explain about multiple document interface.

15. (a) Explain the different sections of data report.

Or

(b) Discuss about data grid control and its properties.

Part C (3 × 10 = 30)

Answer any **three** questions.

16. Explain looping statements with example.

17. Explain the features of menu editor.

18. Explain any five graphics methods with its usage.

19. Explain the usage of common dialogs control with example.

20. Explain the features of visual data manager.

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AFC–7610

Sub. Code
1BCEA1/ 1BCESA1

U.G. DEGREE EXAMINATION, NOVEMBER 2015

Computer Science

Allied – DIGITAL ELECTRONICS

(CBCS – 2011 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. What is the use of Boolean Algebra?

2. What are the universal gates?

3. Write the Boolean equations for the Half Adder outputs.

4. Write the BCD equivalent for the numbers FF_{16} and EE_{16}.

5. What are the Applications of Flip Flops?

6. Write the truth table for D-Flip Flop.

7. What is meant by ECL?

8. What is meant by CMOS?

9. Which principle is used in Resistive Divider?

10. Define “Resolution” of a D/A converter.

Part B $(5 \times 5 = 25)$

Answer **all** questions, choosing either (a) or (b).

11. (a) Do the following conversions

    (i) FF_{16} in to binary equivalent

    (ii) 256_{10} in to octal equivalent

Or

(b) Explain the positive and negative logic.

12. (a) Explain the Magnitude Comparator.

Or

(b) Draw the Logic circuit for 16 to 1 Multiplexer.

13. (a) Explain the JK-master slave Flip Flop.

Or

(b) Describe Reing Counter.

14. (a) Describe the Bipolar Transistor characteristics.

Or

(b) Explain the DTL circuit.

15. (a) Explain the D/A converters in detail.

Or

(b) Explain the Simultenous conversion of A/D converter.

Part C $(3 \times 10 = 30)$

Answer any **three** questions.

16. Draw the Sum of Products logic circuit for the following three input conditions 000, 010, 100, 110.

17. Explain the Decimal Adder.

18. Explain the RS- Flip Flop with truth table and circuit.

19. Explain the TTL circuits in detail.

20. Explain the Dual slope A/D conversion.

AFC–7611

Sub. Code
1BCESA2

U.G. DEGREE EXAMINATION, NOVEMBER 2015

Computer Science — Allied

MICROPROCESSOR AND INTERFACING

(CBCS – 2011 onwards)

Time : Three Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. What are the uses of Index registers of 8086 microprocessor?

2. Write the technical parameters of Address bus, and databus of 8086 microprocessor.

3. Write the syntex for register indirect addressing mode.

4. Write the function of DAS Instruction.

5. Draw the internal architecture of 8255 (PIC).

6. What are the duties of 8259?

7. Write the differences between RAM and ROM.

8. What are the functions of Acoustic coupler?

9. What is meant by MSW?

10. What are the functional parts are available in 80286 microprocessor?

Part B (5 × 5 = 25)

Answer **all** questions, choosing either (a) or (b).

11. (a) Explain the segment registers of 8086 microprocessor.

Or

(b) Write the comparison between 8086 and 8088 microprocessor.

12. (a) Explain the instruction categories of 8086/8088 Microprocessor.

Or

(b) Describe the Assembler directives in detail.

13. (a) Explain the method of Interfacing I/O devices with 8088 microprocessor.

Or

(b) Describe the modes of operation of 8255.

14. (a) Explain about the floppy disk controller.

Or

(b) Explain about the CRT controller.

15. (a) Explain the Register Set of 80286 microprocessor.

Or

(b) Explain the Register Organization of 80386 microprocessor.

Part C (3 × 10 = 30)

Answer any **three** questions.

16. Explain the signal description of 8086 microprocessor.

17. Explain about the stacks of 8086/8088 microprocessor.

18. Describe the features of 74LS373 latch.

19. Describe about the BIOS.

20. Describe the data types supported by 80386 microprocessor.

AFC–7612

Sub. Code
1BCESA3

U.G. DEGREE EXAMINATION, NOVEMBER 2015

Computer Science – Allied

RESOURCE MANAGEMENT TECHNIQUES

(CBCS – 2011 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. Write any two features of OR.

2. Define operation research.

3. Define slack variable.

4. Define optimum basic feasible solution.

5. State the fundamental theorem of duality.

6. Write the symmetric form of primal problem.

7. Write the objective of assignment schedule.

8. Define dummy rows in assignment problem. Give example.

9. Define transportation problem.

10. Define basic feasible solution in transportation problem.

Part B $(5 \times 5 = 25)$

Answer **all** questions choosing either (a) or (b).

11. (a) Explain the O.R modeling.

Or

(b) Discuss the main phases of O.R.

12. (a) Write the procedure for mathematical formulation of L.P.P.

Or

(b) Solve graphically the following L.P.P :

Maximize $z = 3x_1 + 4x_2$

Subject to the constraints

$$4x_1 + 2x_2 \leq 80$$
$$2x_1 + 5x_2 \leq 180$$
$$x_1, x_2 \geq 0$$

13. (a) Write the dual of the L.P.P

Minimize $z = 4x_1 + 6x_2 + 18x_3$

Subject to the constraints

$$x_1 + 3x_2 \geq 3;$$
$$x_2 + 2x_3 \geq 5$$
and $$x_1, x_2, x_3 \geq 0$$

Or

(b) Prove that the dual of the dual is the primal.

14. (a) Solve the assignment problem :

$$\begin{array}{c} \\ \text{I} \\ \text{II} \\ \text{III} \\ \text{IV} \end{array} \begin{array}{c} \begin{array}{cccc} \text{A} & \text{B} & \text{C} & \text{D} \end{array} \\ \begin{pmatrix} 1 & 4 & 6 & 3 \\ 9 & 7 & 10 & 9 \\ 4 & 5 & 11 & 7 \\ 8 & 7 & 8 & 5 \end{pmatrix} \end{array}$$

Or

(b) Solve the travelling salesman problem so as to minimize the cost per cycle.

From To

	A	B	C	D	E
A	–	3	6	2	3
B	3	–	5	2	3
C	6	5	–	6	4
D	2	2	6	–	6
E	3	3	4	6	–

15. (a) Write the procedure for North-west corner rule of transportation problem.

Or

(b) Obtain an initial basic feasible solution to the following transportation problem.

	D	E	F	G	Available
A	11	13	17	14	250
B	16	18	14	10	300
C	21	24	13	10	400
Requirement	200	225	275	250	

Part C (3 × 10 = 30)

Answer any **three** questions.

16. Describe the O.R techniques to solving problem.

17. Use simplex method to solve L.P.P

Maximize $z = x_1 + 2x_2$

Subject to the constraints

$$-x_1 + 2x_2 \leq 8;$$
$$x_1 + 2x_2 \leq 12;$$
$$x_1 - x_2 \leq 3$$
$$x_1, x_2 \geq 0$$

18. Use dual simplex method to solve L.P.P

Maximize $z = 3x_1 - x_2$

Subject to the constraints :

$$x_1 + x_2 \geq 1;$$
$$2x + 3x_2 \geq 2;$$
$$x_1, x_2 \geq 0$$

19. Write the computational procedure for assignment algorithm.

20. Solve the following transportation problem to minimize total cost.

					Available units
	6	1	9	3	70
	11	5	2	8	55
	10	12	4	7	90
Required units	85	35	50	45	

AFC–7613

Sub. Code
1BCESA4

U.G. DEGREE EXAMINATION, NOVEMBER 2015

Computer Science — Allied

COMPUTER ORIENTED NUMERICAL METHODS

(CBCS – 2011 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. Define Bisection method.

2. Define Ill-conditioned equation.

3. Write linear law.

4. Define Eigen values.

5. Write Lagrange's interpolation formula.

6. Define interpolation.

7. Write a first derivative Newton's forward formula.

8. Explain Simpson's rule.

9. Define Runge-Kutta second order method.

10. Define Euler's method.

Part B $(5 \times 5 = 25)$

Answer **all** questions, choosing either (a) or (b).

11. (a) Find a real root of the equation $f(x) = x^3 - x - 1$ by bisection method.

Or

(b) Use the method of iteration to find a positive root between 0 and 1 of the equation $xe^x = 1$.

12. (a) Explain the method of group averages.

Or

(b) Describe the method of least square.

13. (a) The table below gives the values of tan x for $0.10 \le x \le 0.30$.

x :	0.10	0.15	0.20	0.25	0.30
$y = \tan x$:	0.1003	0.1511	0.2027	0.2553	0.3093

Find tan 0.12.

Or

(b) Certain corresponding values of x and $\log_{10} x$ are (300, 2.4771), (304, 2.4829) (305, 2.4843) and (307, 2.4871) find $\log_{10} 301$.

14. (a) From the following table of values of x and y obtain $\frac{dy}{dx}$ and $\frac{d^2y}{dx^2}$ for $x = 1.2$.

x :	1.0	1.2	1.4	1.6	1.8	2.0	2.2
y :	2.7183	3.3201	4.0552	4.9530	6.0496	7.3891	9.0250

Or

(b) Evaluate $I = \int_0^1 \sin \pi x \, dx$ by trapezoidal rule with $h = 0.125$.

15. (a) Find the value of $y(0.1)$ for the differential equation $y'' - xy' - y = 0$ with $y(0) = 1$ and $y'(0) = 0$ use Taylor's series method.

Or

(b) Given $\frac{dy}{dx} = 1 + y^2$ where $y = 0$ when $x = 0$ find $y(0.2)$, $y(0.4)$ and $y(0.6)$ using Runge-Kutta second order formula.

Part C (3 × 10 = 30)

Answer any **three** questions.

16. Use Gauss elimination to solve

$2x + 3y + z = 9,$

$x + 2y + 3z = 6,$

$3x + y + 2z = 8.$

17. Construct a least square quadratic approximation to the function $y(x) = \sin x$ on $[0, \pi/2]$ w.r.t weight function $w(x) = 1$.

18. Explain the Lagranges interpolation formula.

19. Evaluate $I = \int_0^1 \frac{1}{1+x}\, dx$ with $h = 0.5$, 0.25 and 0.125.

20. Given the differential equation $\frac{dy}{dx} = \frac{x^2}{y^2+1}$ with initial condition $y(0) = 0$ use Miline's method to obtain y for $x = 0.25, 0.5$ and 1.0.

AFC–7614

Sub. Code
1BCE1A1

U.G. DEGREE EXAMINATION, NOVEMBER 2015

Computer Science – Allied

OFFICE AUTOMATION (THEORY AND LAB)

(CBCS – 2011 onwards)

Time : 3 Hours Maximum : 60 Marks

Part A (10 × 1.5 = 15)

Answer **all** questions.

1. Write the steps for saving the word document.

2. List out the shortcut commands to find and replace a word in word document.

3. What is the purpose of help command in word document?

4. Write the steps to view a word document in print layout format.

5. Write the steps to create the table in a word document.

6. How to insert a page number in a word document?

7. What is Word Art?

8. Write the steps to open an existing Excel workbook.

9. What is a format painter in Excel?

10. Write steps for filling the color in a cell.

Part B (5 × 3 = 15)

Answer **all** questions choosing either (a) or (b).

11. (a) What are the advantages of word document?

Or

(b) What are the ways to align the text in word document? Explain.

12. (a) How to save and resave a word document?

Or

(b) Write the steps for inserting different shapes in a word document.

13. (a) Write a note on Page setup in a word document.

Or

(b) What are the features of Excel workbook?

14. (a) Write the steps for inserting graphics in a word document.

Or

(b) How to insert a Word Art in a word document? Explain.

15. (a) What are the advantages of using a formula in Excel?

Or

(b) How to edit a data in Excel? Explain.

Part C (3 × 10 = 30)

Answer any **three** questions.

16. How to insert a header and footer in a word document? Give an example with neat diagram.

17. Explain the following commands with suitable example.

    (a) Cut and Paste

    (b) Header and Footer

    (c) Undo and Redo

    (d) Selecting text.

18. Explain various options available in Font dialog box with neat diagram.

19. Describe the various steps to create the Mail Merge in a word document.

20. Explain any ten functions in Excel worksheet.

AFC–7615

Sub. Code
1BCEA2

U.G. DEGREE EXAMINATION, NOVEMBER 2015.

Computer Science

Allied — PRINCIPLES OF INFORMATION TECHNOLOGY

(CBCS – 2011 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. Write the developments in communication technology?

2. What are the applications of computer communication?

3. What is meant by integrated software?

4. What are the applications of web browsers?

5. Write the technical parameters of ISDN lines.

6. What is the use of cable modems?

7. What is meant by file management systems?

8. Write about the diskettes.

9. Name some languages used in internet programming.

10. What are the phases of system analysis and design?

Part B $(5 \times 5 = 25)$

Answer **all** questions choosing either (a) or (b).

11. (a) Explain the revolution in computer communication.

Or

(b) Explain the ethics of information technology.

12. (a) Explain the use of presentation graphics software.

Or

(b) Explain the specilized softwares.

13. (a) Describe about telecomputing.

Or

(b) Describe telephone related services.

14. (a) Explain the types of database organization.

Or

(b) Describe about the magnetic tapes.

15. (a) Explain about the visual programming.

Or

(b) Describe about management information system.

Part C $(3 \times 10 = 30)$

Answer any **three** questions.

16. Explain the digital age at present.

17. Explain the desktop accessories.

18. Explain the on-line information services.

19. Explain the Hierchy and concept of the key field.

20. Explain the steps in programming.

AFC–7616

Sub. Code
1BCEA3

U.G. DEGREE EXAMINATION, NOVEMBER 2015

Computer Science

Allied – PROGRAMMING IN C (THEORY AND LAB)

(CBCS – 2011 onwards)

Time : 3 Hours Maximum : 60 Marks

Part A (10 × 1.5 = 15)

Answer **all** questions.

1. Define Token. Give an example.

2. What is the purpose of loop statement?

3. How will you initialize an one dimensional array?

4. Write any four String handling functions in C.

5. How will you read strings from terminal?

6. What is the main difference between a Structure and Union?

7. Write any Three Pointer expressions.

8. Write the scaling factors for char and double type data.

9. How will you close a file ? Give an example.

10. Define Macro.

Part B $(5 \times 3 = 15)$

Answer **all** questions, choosing either (a) or (b).

11. (a) Explain the fundamental Data types in C with example.

Or

(b) Explain the precedence of arithmetic operators in C with example.

12. (a) Write a C program to find the sum of odd numbers in an array containing 20 elements.

Or

(b) Write a C program to reverse the given string.

13. (a) Explain Recursion with an example.

Or

(b) Explain with an example how will you create and initialize a union.

14. (a) Write a program to calculate the area of a triangle using pointers.

Or

(b) Explain the advantages of Pointer.

15. (a) Explain the various File opening modes with example.

Or

(b) Explain any five compiler control directives in C.

Part C $(3 \times 10 = 30)$

Answer any **three** questions.

16. Write a C program to find the sum of the numbers that are divisible by 9 and not divisible by 15, between 100 and 500.

17. Write a C program to multiply two matrices. Check the comtability.

18. Write a C program to print the mark statement of the students using array of structures. (Assume your own data)

19. Explain Pointers and structures with a program.

20. Write a C program to print the address of the employees using file. (assume your own data)

AFC–7465

Sub. Code
1BCE1C1

B.Sc. DEGREE EXAMINATION, NOVEMBER 2015

First Semester

Computer Science

PROGRAMMING IN 'C' AND DATA STRUCTURE

(CBCS – 2011 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. What is conditional operator?

2. What is use of break and continue statement?

3. What is an array?

4. What is dynamic memory allocation?

5. Define self referential structure.

6. Define Enum data type.

7. Define Linked List.

8. What is the disadvantage of linear representation of queue?

9. What is degree of a node?

10. Define Tree.

Part B $(5 \times 5 = 25)$

Answer **all** questions choosing either (a) or (b).

11. (a) Compare switch and if . . . else if statement.

Or

(b) Explain any three type of operators supported in C?

12. (a) How do declare and access multi dimensional arrays?

Or

(b) Explain array of pointers with an example.

13. (a) Explain any three operations used in file.

Or

(b) Write 'C' program to create a data file.

14. (a) Explain the push and pop operations in stack.

Or

(b) How is queue represented?

15. (a) How do you represent Binary Tree using an array?

Or

(b) How do represent Binary Tree using Linked list.

Part C $(3 \times 10 = 30)$

Answer any **three** questions.

16. Discuss the storage classes in 'C'.

17. How do pass function to other function. Give an example.

18. Explain Insertion and deletion operations in Linked List.

19. Explain Applications of Stack.

20. Discuss Applications of Trees.